

Theoretical Computer Science, exam of 2025.09.02

SURNAME _____ NAME _____ PERSONAL CODE _____

Total available time: 1h 30'. You can use books and paper notes during the exam, while electronic devices are prohibited. **You must write the solutions on this paper alone.** Each correct exercise is valued 11 points. Multichance students do not need to answer to Exercises 2.2 and 3.2: bar here in that case []

Exercise 1

Consider the following two languages over the alphabet $\Sigma = \{a, b\}$, and provide a recognizer of minimal power for each language. Given a string $s \in \Sigma^*$, $|s|_a$ denotes the number of occurrences of a in it, and $|s|_b$ denotes the number of occurrences of b .

1) The language L_1 of strings such that, for every prefix p of the string, the absolute value of the difference between $|p|_a$ and $|p|_b$ does not exceed 2.

2) The language L_2 of strings $s \in \Sigma^*$ such that the absolute value of the difference between $|s|_a$ and $|s|_b$ is exactly 1.

Exercise 2

Let $\mathbf{F}$ be the set $\{f\}$, with f defined as follows:

$f(x) = \mathbf{if}$ the x -th machine in Gödel's enumeration computes a partial function **then** 0 **else** 1.

1) Is it decidable to establish whether a generic Turing machine computes a function contained in $\mathbf{F}$?

2) Let $\mathbf{G}$ be the set of functions $\{f, g\}$, with $\forall x, g(x) = \perp$ and f defined as above. Is it decidable to establish whether a generic Turing machine computes a function in $\mathbf{G}$?

Exercise 3

Consider the language L over the alphabet $\Sigma = \{0, 1, 2\}$ consisting of strings s of length $|s| = l, l \in \mathbb{N}$, such that there exists a substring of minimum length $\lceil l/2 \rceil$ in which both the number of 1s and the number of 0s are even.

1) Describe a k -tape Turing Machine that recognizes the language L , and evaluate its space and time complexity.

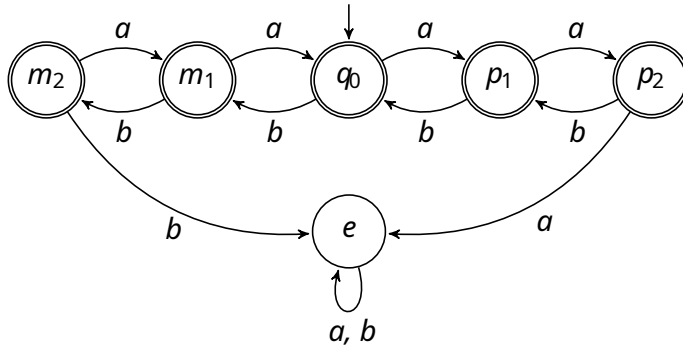
2) How does the recognition strategy change, and consequently how do the space and time complexities change, if the constraint on the minimum length is removed?

space available for solutions

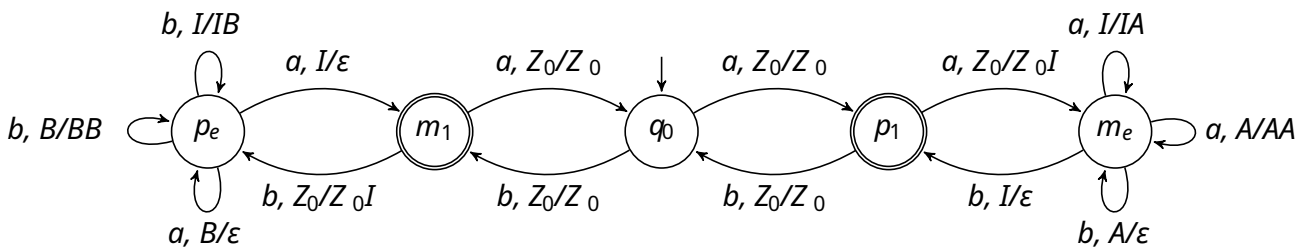
Solutions

EX 1

1) The language L_1 is regular. In fact, it is sufficient to compute the absolute difference by means of a finite automaton that keeps track of it only insofar as it lies between -2 and 2 as the input characters are read.



2) The language L_2 is context-free: the possible values of the difference between $|s|_a$ and $|s|_b$ that need to be tracked are unbounded.



EX 2

1) Decidable (and therefore also semi-decidable) and also decided negatively, since no machine computes f (the totality problem of a function is notoriously undecidable). Therefore, $\mathbf{F}$ contains no computable function.

2) In this case the problem is not decidable by Rice's theorem. We are in a nontrivial case of the theorem since there exists a computable function in the set $\mathbf{G}$ (the function g itself) and one not in the set (for example, the function $h(x) = x$).

Moreover, the problem is not even semi-decidable because the only computable function in the set $\mathbf{G}$ is the everywhere-undefined function g : establishing non-equivalence with g is semi-decidable (by diagonal enumeration: it is enough to observe an input for which the machine halts to establish non-equivalence). Therefore, the problem in question, which is the complement of that, cannot be semi-decidable.

EX 3

1) The Turing Machine first computes, in one pass, $l/2$ in unary on the first tape (writing one symbol every two input symbols). After this, it repeats the following procedure, keeping track of the parity of the number of 0s and 1s through the states of the control unit, and using a second tape as a unary counter for the number of letters to skip from the beginning of the input (initially set to 0):

(a) Return to the beginning of the input, skip a number of letters equal to the value stored on the second tape, and set the parity values to (even, even).

(b) Scan a number of input letters equal to the unary value stored on the first tape, updating the parity values. If the input ends before this, halt and reject the input string.

(c) Repeat the following sub-procedure until the end of the input:

- If both parity counters are “even,” accept.
- Read one more input symbol and update the parity counters.

(d) If the sub-procedure terminates without acceptance, increment the counter on the second tape by adding one symbol, and return to step (a).

The time complexity of the procedure is quadratic in l , while the space complexity is linear in l .

2) Any string in Σ^* admits the empty string as a substring, which contains an even number of 0s and 1s. Therefore, $L = \Sigma^*$, and recognizing whether a string belongs to L requires only constant time and constant space.